

Experiment:10

Illustrate the concept of inter-process communication using message queue with a c program

Code:-

```
#include <stdio.h>
#include <stdlib.h>
#include <string.h>
#include <unistd.h>
#include <sys/types.h>
#include <sys/ipc.h>
#include <sys/msg.h>

struct message
{
    long msg_type;
    char msg_text[100];
};

int main()
{
    key_t key = ftok("msgqfile", 65);

    if (key == -1)
    {
        perror("ftok");
        exit(EXIT_FAILURE);
    }

    int msgid = msgget(key, IPC_CREAT | 0666);
```

```
if (msgid == -1)
{
    perror("msgget");
    exit(EXIT_FAILURE);
}

struct message msg;

msg.msg_type = 1;

strcpy(msg.msg_text, "Hello, message queue!");

if (msgsnd(msgid, &msg, sizeof(msg.msg_text), 0) == -1)
{
    perror("msgsnd");
    exit(EXIT_FAILURE);
}

printf("Producer: Data sent to message queue: %s\n", msg.msg_text);

if (msgrcv(msgid, &msg, sizeof(msg.msg_text), 1, 0) == -1)
{
    perror("msgrcv");
    exit(EXIT_FAILURE);
}

printf("Consumer: Data received from message queue: %s\n", msg.msg_text);

if (msgctl(msgid, IPC_RMID, NULL) == -1)
{
    perror("msgctl");
    exit(EXIT_FAILURE);
}

return 0;
```

```
}
```

Output:

```
ftok: No such file or directory

...Program finished with exit code 1
Press ENTER to exit console.
```